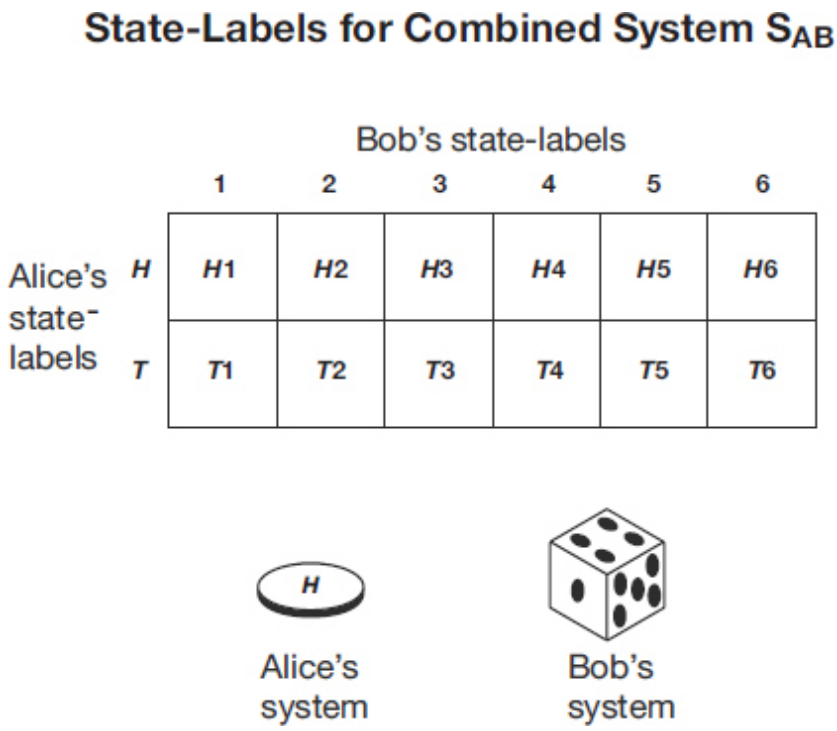


Figure 6.1: The basis states of the composite system S_{AB}, shown as a table. Across the top are the state-labels for Bob's die. The state-labels for Alice's coin are shown on the left. The state-labels for the combined system are the table entries. Each combined state-label shows the state of each of the two subsystems. For example, the state-label *H*4 denotes a state in which Alice's coin shows *H* and Bob's die shows 4.

There are various ways to represent these states symbolically. We could represent the *H*4 state using explicit notation, as |*H*}⊗|4> or |*H*}|4>. Usually, it's more convenient to use the composite notation |*H*4>. This emphasizes that we're talking about a single state with a two-part label. The left half labels Alice's subsystem, and the right half labels Bob's. The explicit and composite notations both have the same meaning—they refer to the same state.

Once the basis vectors are listed—in this case, twelve of them—we can combine them linearly to form arbitrary superpositions. Thus, the tensor product space in this case is twelve-dimensional. A superposition of two of these basis vectors might look like

$$\alpha_{h3}|H3\rangle + \alpha_{t4}|T4\rangle.$$